

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1050

Blue Ravine-School 230 kV
April 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1050. The Interconnection Customer proposes a 250.4 MW solar photovoltaic facility interconnecting at the Blue Ravine-School 230 kV substation in the Indiana area. The request was received on August 30, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 333 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,331,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BONNY EAGLE-RED LION 100 KV line	BLUE RAVINE-SCHOOL-THOMAS HILL STATION SUBSTATION 230 KV	126.3	141	5.9
BONNY EAGLE-OAK PLAINS 100 KV Line	BLUE RAVINE-SCHOOL-THOMAS HILL STA SUBSTATION 230 KV line	119.4	137	5.9
THOMAS HILL STATION 500/230kV Auto	DOOVILLE-MONTPELIER ELECTRIC GENERATING STATION 230 KV	116.4	1,009	37.6
TIMBER SWITCH-MUSKOGEE 230 KV	SWAN VALLEY-TREGO 500 KV L	106.1	731	6.3
WEST MELROSE-TIMBER SWITCH 230 KV	SWAN VLY-TREGO 500 KV	105.0	770	6.2
BLUE RAVINE-SCHOOL-DESOTO 230 KV L	BLUE RAVINE-SCHOOL-HENRY COUNTY 230 KV	103.0	1,092	41.7
15TH AVENUE S-LUCENT 230 KV Line	OMNI-HACKBERRY SUBSTATION-NORWICH 500 KV L	102.7	921	6.0
15TH AVENUE SOUTH-NORWICH 230 KV line	OMNI-HACKBERRY-NORWICH 500 KV	102.4	939	6.0
BLUE RAVIE-SCHOOL-HENRY COUNTY 230 KV line	BLUE RAVINE-SCHOOL-DESOTO SUBSTATION 230 KV	102.2	1,195	48.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BONNY EAGLE-RED LION 100 KV L	Rebuild line with higher-capacity conductor	\$5,874,000	14% (\$840,000)
BONNY EAGLE-OAK PLAINS 100 KV Line	Rebuild line with higher-capacity conductor	\$12,523,000	14% (\$1,792,000)
THOMAS HIL STA 500/230 kV TX	Replace transformer with higher-rated unit	\$24,904,000	6% (\$1,412,000)
TIMBER SWITCH-MUSKOGEE 230 KV	Reconductor line	\$5,288,000	3% (\$177,000)
TIMBE SWITCH-W MELROSE 230 KV	Reconductor line	\$50,825,000	3% (\$1,708,000)
BLUE RAVINE-SCHOOL-DESOTO 230 KV	Reconductor line	\$13,507,000	10% (\$1,406,000)
15TH AVENUE S-LUCENT 230 KV Line	Reconductor line	\$4,285,000	2% (\$86,000)
15TH AVENUE SOUTH-NORWICH 230 KV line	Reconductor line	\$5,343,000	2% (\$107,000)
BLUE RAVIE-SCHOOL-HENRY COUNTY 230 KV line	Reconductor line	\$16,007,000	11% (\$1,803,000)
	Total allocated to project		\$9,331,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.